Adroit Technologies Protocol Driver

Name	PolyComp Sign Ethernet TCP/IP
DLL	POLY_ETH

COPYRIGHT

Advanced Worx 112 (Pty) Ltd
20 Waterford Office Park
189 Witkoppen Road (c/o Waterford Drive)
Fourways
South Africa

https://adroittech.co.za/

Contents

1. Overview

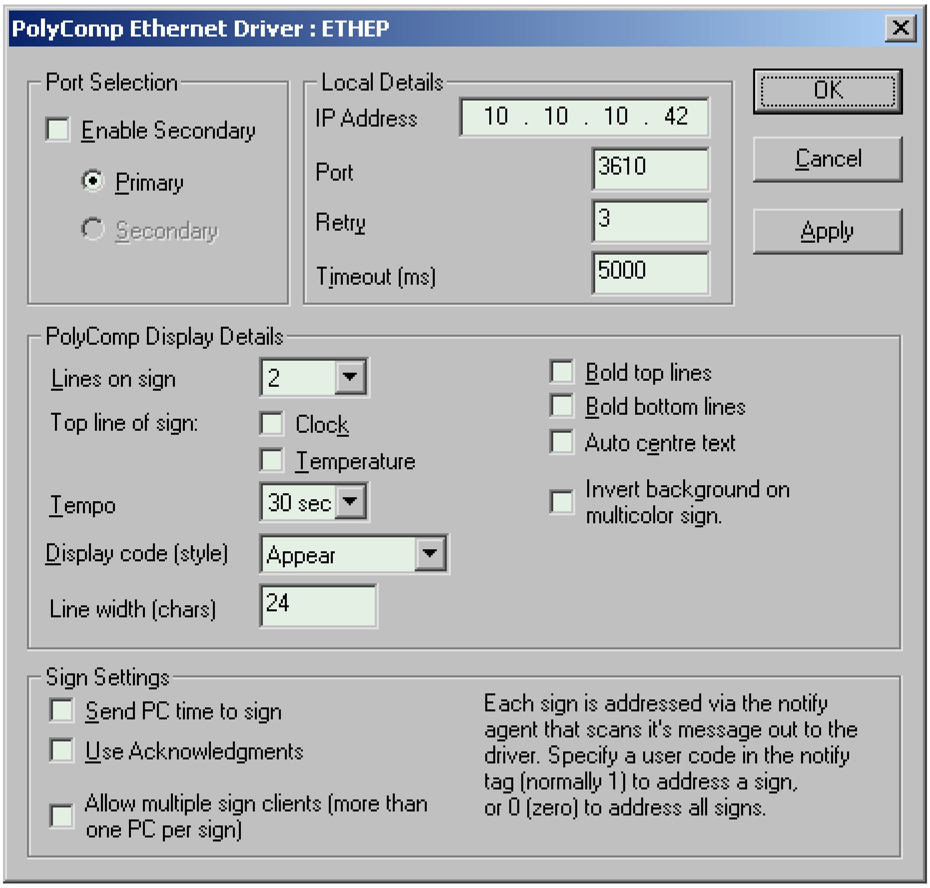
Adroit supports communication with PolyComp GTX signs on ethernet. Communication is via an Ether-PAD box installed into the sign.

2. Wiring

Ethernet, the Ether-Pad box installation includes an external RJ-45 connector. For more on the E-88 Ethernet-serial conversion box please refer to the manufacturer's documentation.

3. Device Configuration

3.1. Device Configuration



3.1.1. Port Selection

Enable Secondary – Enables secondary configuration for dual redundant Controllers. This setting is disabled for this driver.

Primary / Secondary – Select the primary or secondary button depending on which connection you wish to configure.

3.1.2. PolyComp Display Details

Lines on Sign – The number of lines on the sign, can be 1-127

Top line of sign **:Clock –** Displays the date and time on the top line
:Temperature - Displays the temperature (in bold)

Tempo – The tempo or duration to display the message for, affects the use of certain styles.

Bold top line – Top 2 lines of sign in bold

Bold bottom line - Lines 3 & 4 in bold

Auto centre text – Text centres itself on the sign

Display code (style)- Controls the way that text appears on the display.
Line width Specify the line width (maximum of 40) in characters wide that the display is. This will only affect datascope text displayed.

3.1.3. Sign Settings

Use acknowledgments – The driver should request and wait for the ACK from the sign. If this option is on, and the sign must acknowledge in order for the driver to be able to determine if the communication is working. *This option may not work on most signs, if the RS232 transmit from the sign is not connected.*

Send PC time to sign – The driver will send the local time to the Sign each time a message is sent.

Allow multiple sign clients – The driver will disconnect and reconnect with each message sent, this will allow use in a cluster or situation where 2 devices control the sign content.

3.1.4. Local Details

IP Address – TCP/IP address of the sign.

Port – The port used at this IP address to communicate.

Retry - The number of times that Adroit will try to gain communication with a PLC before failing a transaction.

Timeout - The time that Adroit will wait for a valid response from a PLC before failing a transaction (Read or Write).

PLEASE NOTE

The Clock and temperature options (displayed on the top line only) invoke the feature if present in the sign. If the sign does not have an internal clock (requires time sent from PC) and temperature sensor, these options have no effect.

Apply Button – Use this button to save the settings without leaving the dialog, you may use the dialog to customise the sign appearance (any setting in the sign details group) and then send a message to the sign straight away from Adroit to see the effects of the selections.

4. Supported Addresses

4.1. Addresses

The driver will scan a string slot using an address mnemonic made up of an alphabetic and numeric characters, e.g. : "FRED001"
The string will be converted to upper case and is used by the driver to identify each message apart.

PLEASE NOTE
THE SCAN-RATE MUST BE ZERO!

4.2. Scanning

You may use a notify agent to scan, put a Sign address of 1 into the user code of the notify agent, and then scan the rawValue slot to the driver. Each notify agent can thus address a sign uniquely. The default address for a sign (since usually only one sign will be on the Ether-Pad) is 1.

5. Protocol Definition and Considerations

5.1. Message Details

You must manually address a sign when not using the Notify agent by inserting the station/sign number in at the beginning of the string followed by a comma, e.g.: "5,East-dam level high", will address the sign #5 with the message "East-dam level high".

Control codes may be sent to the sign using the character '` ', same key as the tilde '~ ' on the keyboard, the driver converts this to the control code 28, thus the message "Call `FAdroit" Would display with only "Adroit" flashing.

5.1.1. Control Codes are:

F : Flash Characters
E : Enlarge Characters
R : Change Colour to Red *(Color signs only!)*
G : Change Colour To Green *(Color signs only!)*
Y : Change Colour to Yellow *(Color signs only!)*
M : Multicolour – i.e.: Top Red Center Yellow Bottom Green and applies to everything after the `M *(Color signs only!)*
D : Return To Default Setting – i.e.: Normal Red Letters Not Flashing. (All the above settings are cleared)
T : Turn background on (applies to entire message, and also to subsequent messages)
t : Turn background off (applies to entire message, and also to subsequent messages)

5.1.2. Background Colors.

The sign background can be enabled by sending a `T to the sign, this is a toggle that effects all subsequent messages. Using `t (lower-case T) turns the background off. The background setting code may appear anywhere in the message (it gets removed) and will affect the entire message. Background for certain colors may not be permissible depending on the sign used.

5.1.3. Style Notes

When using "scroll" style, the 1st line displayed in multi-line displays will not scroll, use the BOLD style or control code to scroll all the text.

5.1.4. Longer Messages

Long messages up to 180 characters are possible using the continuation character ` (same as for control codes) at the end of a message. All messages sent to the same scan address (E.g. NAME 0) are appended up to the above limit. The last message must not end in the control code; this triggers the sending of the built-up long message.
E.g. Scan a notify agent from NAME 1, and then add the text "Hello world`", and send it, then add " how are you today`" and lastly add "?". All 3 messages will be concatenated and sent as one. "Hello world how are you today?"

Any characters over the limit are discarded (without notice) at the time the message is built up for sending to the sign.

5.1.5. Using Time and Temperature

You must allow for the 1st line of the sign being swallowed up and will have to pad the message sent (add 25 characters) to the beginning of the message sent. This does not work is the display is not fitted with the relevant option.

5.1.6. Actual Sign Appearance

The display may not always appear as expected; some options are incompatible with others, and also with certain signs capabilities. You might try using the `D to return the current text to default, and `t to turn backgrounds off. Notice that the display can take up to 0.5 seconds to display the new message.

5.2. Message Protocol

Communications are TCP/IP protocol, via the Pandora box. Because the box only allows one socket connected at a time, simultaneous access to the sign from different machines or applications is only possible through application disconnecting first.

Protocol is unchanged from the serial protocol, except in that for fault-tolerant operation, acknowledgements are required.

The LRC value is the 8 bit XOR of all bytes in a message including the SYNC and the EOT characters. This section documents the protocol, which has been condensed here.

SYNC	Lines	Address	ETX	Msg flag	Page	Data	EOT	CRC
x00	Nn	aaa	x03	f	ppp	data	x04	x??

ACK
x06

Nn = number of text lines on display unit.
Aaa = unit address, 1-127, address 0 = all units
F = Flag, see **note 1**
Ppp = Page number, "000" to "999"
Data = Data, may be the time or a message, see **note 2**
CRC = Cyclic redundancy check, calculated

Note 1:
Message status flag.

Bit 0	0
Bit 1	1= Interrupt mode, 0=normal
Bit 2	1= More pages to come
Bit 3	1= Ack requested
Bit 4	1= Schedule information contained in data
Bit 5	0
Bit 6	1
Bit 7	1

Note 2:

If the page is page 0 (scheduled flag=on) the layout is as follows:

Byte	Designation
9	Hours (MSB)
10	Hours (LSB)
11	Minutes (MSB)
12	Minutes (LSB)
13	Seconds (MSB)
14	Seconds (LSB)
15	Day of month (MSB)
16	Day of month (LSB)
17	Month (MSB)
18	Month (LSB)
19	Weekday (MSB) always = 0
20	Weekday (LSB) Sunday=1 …

If any other page has the scheduled flag the layout is as follows:

Byte	Designation	
9	Start day (MSB)	
10	Start day (LSB)	
11	Start month (MSB)	
12	Start month (LSB)	
13	Stop day (MSB)	
14	Stop day (LSB)	
15	Stop month (MSB)	
16	Stop month (LSB)	
17	Start hours (MSB)	
18	Start hours (LSB)	
19	Start minutes (MSB)	
20	Start minutes (LSB)	
21	Stop hours (MSB)	
22	Stop hours (LSB)	
23	Stop minutes (MSB)	
24	Stop minutes (LSB)	
25	Weekday status	
	Bit 0	1= display on Sunday
	Bit 1	1= display on Monday
	Bit 2	1= display on Tuesday
	Bit 3	1= display on Wednesday
	Bit 4	1= display on Thursday
	Bit 5	1= display on Friday
	Bit 6	1= display on Saturday
	Bit 7	1

Message portion of data (does not apply to page 0)

Tempo	Function	State	Message text
Note3	Note4	Note5	Any message text*

* Control codes may be imbedded in the message when using multi-colour signs.

Note 3:

Bit 0-3 Tempo -

Value	1	2	3	4	5	6	7	8	9
Time (s)	2	5	10	20	30	45	60	90	120

Bit 4-5 Display status bits -
Both off = timer mode (schedule)
0,1 = always on
1,0 = always off
1,1 = invalid

Bit 6-7 Both 1

Note 4:

Bit 0-3				
0	1	2	3	Function -
0	0	0	0	0=auto
0	0	0	1	1=Appear
0	0	1	0	2=Wide
0	0	1	1	3=Open
0	1	0	0	4=Lock
0	1	0	1	5=Rotate
0	1	1	0	6=Right
0	1	1	1	7=Left
1	0	0	0	8=Roll up
1	0	0	1	9=Roll down
1	0	1	0	10=Ping pong
1	0	1	1	11=Fill up
1	1	0	0	12=Paint
1	1	0	1	13=Fade in
1	1	1	0	14=Jump
1	1	1	1	15=Slide
Bit 4				Display clock on top line
Bit 5				Display temperature on top line
Bit 6-7				Both 1

Note 5:

Bit 0	1= (top 2) lines 1&2 BOLD
Bit 1	1= lines 3&4 BOLD (driver is 0)
Bit 2	1= lines 5&6 BOLD (driver is 0)
Bit 3	1= lines 7&8 BOLD (driver is 0)
Bit 4	1= Auto-centre (driver is 0)
Bit 5	1= Foreign language, 0=English
Bit 6	1= background
Bit 7	1= Reserved (driver is 1)

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback (adroittechnologiesautomation.com)

6.3. Troubleshooting

Q: The sign regularly fails to communicate properly because the driver retries, but still displays the intended message.
A: ensure that the timeout is long enough (about 3 seconds), and restart. A good timeout value will also be dependent on the animation settings chosen; longer animations run slower.

Q: The exact text I intended is displayed, but the colours are wrong.
A: Some color combinations are invalid, try different combinations to find ones that work for your sign.

Q: I cannot communicate at all.
A: Try firstly to ping the display unit's I/P address, if this works, ensure that no other computers are connected to the sign, the sign only allows one connection computer at a time. Lastly, ensure that the port selected is correct.

Q: I cannot get the sign to communicate, but I do not know it's settings.
A: Using HyperTerminal (9600, 8 none 1,no handshaking) connect a serial cable (2to3 and 3to2 with ground; pin 5 through) to the configuration port. Turn the sign off, and then on, you will be prompted for a password xxx. Follow the menus to see the sign settings, and then follow the "B"ack menu to get out and re-boot the Ether-PAD.

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1		K. Robinson	Changes to format, updated image resources.	D. Jordaan
2.0				
3.0				

7.2. DLL Revision

Rev.	Date	Changes
	4-May-99	Initial revision
	10-May-99	Added the 4 missing styles & page #s.
	30-Jun-2002	Ethernet version documented
	11/13-Mar-02	Background colors added, RS232 stuff fixed. Screenshot updated. Longer messages support added; protocol fixed up. Troubleshooter added.
	12-Aug-05	Notes on how clock and temperature actually work added. Datascope enhancement of sign-text preview added.
11	24-May-2021	Added cluster awareness.